MSCA POSTDOCTORAL FELLOWSHIP

Horizon Europe — Marie Sklodowska-Curie Actions

Adaptive Urban Microbiomes: Engineering Bioresponsive Building Surfaces for Climate-Resilient Cities

AUMICA

Field	Value
Fellowship Type	European
Call Identifier	HORIZON-MSCA-2025-PF-01
Duration	24 months
Researcher	Dr. Elena Vasquez
Host Organisation	Politecnico di Milano, Department of Architecture, Built Environment and Construction Engineering (ABC), Italy
Supervisor	Prof. Marco Bellini
Research Field	Environmental Engineering / Microbiology (ENG panel)

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Abstract

AUMICA — Adaptive Urban Microbiomes: Engineering Bioresponsive Building Surfaces for Climate-Resilient Cities — is a 24-month European MSCA Postdoctoral Fellowship conducted by Dr. Elena Vasquez at Politecnico di Milano (Italy), hosted by the Bio-Architecture Laboratory under the supervision of Prof. Marco Bellini. The project pioneers a paradigm shift in bio-integrated building materials: from passive biological colonisation to actively designed microbial ecosystems that deliver measurable environmental services. AUMICA has four interconnected objectives: (1) generating the first continental-scale metagenomic atlas of building-surface microbiomes across five European climate zones; (2) engineering three CRISPR-designed bioresponsive coatings for self-cleaning, air purification, and thermal regulation; (3) validating these coatings on a real living building facade testbed over 12 months; and (4) releasing MicroBuild, an open-source computational model predicting microbiome-material interactions under climate change scenarios. The project bridges environmental microbiology, synthetic biology, materials science, and architectural design — disciplines that have not previously been integrated at this scale and depth. AUMICA will produce three high-impact publications, one patent application with Holcim Innovation Center, a publicly available metagenomic atlas, and a freely available computational toolkit for architects and materials scientists. For Dr. Vasquez, the fellowship constitutes a strategic career transformation toward a tenure-track position in bio-integrated architecture, combining her existing microbiology expertise with new competences in materials science, architectural prototyping, and industry R&D. AUMICA directly addresses the European Green Deal priorities of climate-resilient built environments and circular construction, and contributes to the MSCA goals of researcher mobility, interdisciplinary training, and knowledge transfer between academic and industrial sectors.
Keywords: urban microbiome, bioresponsive building materials, CRISPR biofilm engineering, bio-integrated architecture, environmental metagenomics, climate-resilient cities, synthetic biology, living building facades, MicroBuild, sustainable construction

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PART B1

SECTION 1 — EXCELLENCE

1.1 Quality and Pertinence of the Research Objectives

The built environment represents one of the least explored frontiers in microbiome science. Buildings are not sterile entities: their surfaces host complex microbial communities shaped by climate, material composition, urban density, and human activity. Yet contemporary bio-integrated building research has consistently treated microorganisms as passive agents — colonisers to be tolerated or excluded — rather than as functional partners in building performance. AUMICA challenges this paradigm directly. The project proposes to design building surfaces as engineered microbial ecosystems in which synthetic microbial consortia actively respond to environmental stimuli — heat stress, atmospheric pollution, and humidity fluctuation — delivering measurable services including self-cleaning, air purification, and thermal regulation. This positions AUMICA at the intersection of environmental metagenomics, synthetic biology, materials science, and bio-integrated architecture, an interdisciplinary space that remains largely uncharted at the applied scale.

The state of the art in bio-integrated materials includes self-healing concrete relying on Bacillus-based calcium carbonate precipitation (Jonkers et al., 2010), bioreceptive facade panels designed to support macrophyte growth (Kukec et al., 2019), and recent proof-of-concept studies on engineered biofilms for atmospheric CO2 capture. However, none of these approaches has characterised the native microbiome ecology of building surfaces across diverse European climate zones, nor has any study employed CRISPR-based consortium engineering to create coatings that dynamically adapt their functional outputs to real-time environmental signals. AUMICA goes beyond this state of the art in three distinct ways: (i) it generates the first continental-scale metagenomic atlas of building-surface microbiomes, providing the ecological foundation for rational consortium design; (ii) it employs CRISPR-guided synthetic biology to engineer consortia with defined, switchable functional profiles; and (iii) it validates engineered coatings under real-world conditions on a living building facade testbed over a full 12-month cycle, including seasonal extremes.

The project is structured around four specific, measurable objectives: (O1) Map and characterise microbial communities on building surfaces across five European climate zones — oceanic, Mediterranean, continental, semi-arid, and alpine — using shotgun metagenomics combined with culturomics, producing a publicly accessible European Building Microbiome Atlas; (O2) Design and engineer three bioresponsive surface coatings — a self-cleaning photocatalytic-microbial hybrid, an air-purifying NOx-metabolising consortium coating, and a thermal-buffering evapotranspiration coating — optimised for compatibility with cement-based substrates; (O3) Validate the three coatings on Politecnico di Milano's living building facade testbed under natural environmental conditions over 12 consecutive months, quantifying performance against defined metrics (contact angle reduction, NOx removal efficiency, surface temperature differential); and (O4) Develop and publicly release MicroBuild v1.0, an open-source agent-based computational model predicting microbiome-material interactions under current and projected climate change scenarios to 2050. These objectives are ambitious in their scope and interconnection, yet grounded in demonstrated feasibility through Dr. Vasquez's prior publications and the host's established infrastructure.

1.2 Soundness of the Proposed Methodology

AUMICA employs a four-stage methodological pipeline that integrates field ecology, synthetic biology, materials engineering, and computational modelling.

Stage 1 — Microbiome characterisation (WP1, M1–M8): Field campaigns will be conducted at ten building sites per climate zone (50 sites total), selected to represent variation in building age,

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material type (concrete, brick, render, stone), facade orientation, and urban context. Surface swab and tape-strip samples will be collected following a standardised protocol adapted from the Human Microbiome Project sampling framework. Shotgun metagenomic sequencing will be performed at 10 Gb per sample using Illumina NovaSeq 6000, with bioinformatic processing via the MGSA pipeline (alignment, binning, functional annotation using KEGG and COG databases). Culturomics will complement metagenomics for the recovery of low-abundance taxa with functional relevance. The resulting dataset — approximately 500 metagenomes — will constitute the European Building Microbiome Atlas, deposited in the European Nucleotide Archive under open access.

Stage 2 — Consortium engineering and coating formulation (WP2, M6–M16): Three target consortia will be designed using a design-build-test-learn cycle. CRISPR-Cas12a will be employed to introduce functional genetic modules into chassis organisms selected from the Atlas for their surface-compatibility and robustness. Coating formulations will use biocompatible cement-polymer matrices developed in collaboration with Holcim Innovation Center during the three-month secondment (M9–M11). Climate chamber testing at Politecnico di Milano's ABC laboratory will evaluate coating stability across a matrix of temperature (5–45°C), relative humidity (20–95%), UV irradiance, and simulated acid rain exposure. Accelerated aging protocols (500-hour UV-condensation cycles per ISO 16474) will precede outdoor deployment.

Stage 3 — Living facade validation (WP3, M14–M24): The three coatings will be applied to designated panels of Politecnico di Milano's living building facade testbed — a purpose-built outdoor experimental facility on the ABC department building. Monitoring will include weekly microbiome profiling (16S rRNA and ITS amplicon sequencing), continuous environmental sensor data (temperature, humidity, particulate matter), and monthly functional assays. This constitutes the first real-world, multi-season validation of CRISPR-engineered bioresponsive coatings on an inhabited building.

Stage 4 — Computational modelling (WP4, M4–M22): MicroBuild will be built on a hybrid agent-based / ordinary differential equation framework, incorporating microbial growth kinetics, substrate diffusion, climate forcing variables, and material degradation functions. The model will be calibrated iteratively against WP1 and WP3 empirical data. Scenario simulations will project coating performance under RCP 4.5 and RCP 8.5 climate trajectories to 2050 for five European cities.

Interdisciplinary integration is structural to AUMICA. The project bridges environmental microbiology, synthetic biology, construction materials science, computational biology, and architectural design. Dr. Vasquez's metagenomics and CRISPR expertise is complemented by Prof. Bellini's group's architecture and materials science competence, and by Holcim's industrial formulation knowledge. This tripartite interdisciplinarity is not cosmetic — each discipline contributes essential and non-substitutable inputs to the research pipeline.

Gender and diversity dimension: AUMICA incorporates an explicit gender-analytical lens in Stage 1 sampling design. Building surface microbiome composition may be influenced by the gendered patterns of urban space use: facades proximate to care facilities, schools, and domestic spaces experience different human microbiome inputs than those adjacent to industrial or high-traffic zones. Sampling metadata will include building function and urban zoning classifications, enabling stratified analysis of microbiome variation by gendered spatial use patterns. This follows the European Institute for Gender Equality's framework for integrating gender analysis into environmental science.

Open science: All metagenomic raw data will be deposited in the European Nucleotide Archive (ENA) under FAIR principles (Findable, Accessible, Interoperable, Reusable) with persistent DOI identifiers

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assigned at submission, no later than the time of associated publication. The MicroBuild computational model will be developed openly on GitHub under MIT licence from M4, enabling community contributions during development. A Data Management Plan will be submitted as Deliverable D5.1 by M6, compliant with Horizon Europe requirements. All peer-reviewed publications will be submitted to open-access journals or posted as preprints on bioRxiv/EarthArXiv prior to journal submission. The project will not generate personal data.

1.3 Quality of Supervision, Training and Two-Way Transfer of Knowledge

Prof. Marco Bellini is a Full Professor of Sustainable Building Technologies at Politecnico di Milano and the founding director of the Bio-Architecture Laboratory (BAL) within the ABC department. His research portfolio spans bio-integrated facade systems, responsive building materials, and circular construction, with over 120 peer-reviewed publications and coordinating roles in three Horizon Europe projects (BioFacade-H2020, CircularBuild, and GreenSkin). He has supervised 18 PhD students to completion and currently leads a group of 15 researchers. His track record demonstrates both scientific leadership in bio-integrated architecture and a sustained commitment to cross-disciplinary collaboration. Critically, Prof. Bellini's group has identified microbial ecology as the key missing dimension in their bio-integrated materials programme, making Dr. Vasquez's profile an exact strategic fit. A formal Letter of Commitment from Prof. Bellini and Politecnico di Milano accompanies this proposal.

The training programme for Dr. Vasquez is structured as a Personalised Career Development Plan (PCDP) jointly designed with Prof. Bellini, targeting three skill domains: (i) Technical research skills — advanced materials characterisation (X-ray diffraction, SEM-EDS, confocal microscopy) through supervised laboratory training in the ABC materials testing suite; biocompatible coating formulation principles through the Holcim secondment; and architectural prototyping methodology through BAL design workshops. (ii) Professional and leadership skills — Project Management for Researchers (MSCA-endorsed online course, M2); Research Data Management and FAIR Principles (Politecnico di Milano Graduate School course, M3); and supervision of one Master's student assigned to WP1 data analysis (M6–M18). (iii) Transferable and communication skills — Science Communication for Non-Specialist Audiences (workshop, M4); Italian language proficiency (B2 level course, M1–M12); and academic writing mentorship from Prof. Bellini's group. The PCDP will be formalised as Deliverable D5.2 by M3 and reviewed at M12.

The two-way transfer of knowledge is substantive and directional. From Dr. Vasquez to the host: Dr. Vasquez introduces expertise in shotgun metagenomics, bioinformatic pipeline development, and CRISPR-based microbial engineering that does not currently exist within Prof. Bellini's group. She will deliver two internal seminars (M3 and M15) on environmental metagenomics and CRISPR applications for the BAL research group, co-supervise a PhD student whose project intersects with WP2 consortium engineering, and provide hands-on training in metagenomic data analysis to two early-stage researchers in the group. These contributions will permanently expand the methodological repertoire of the BAL, positioning it to lead the emerging field of microbially-informed bio-integrated design. From host to researcher: Prof. Bellini's group will transfer expertise in architectural design thinking, building physics, materials performance assessment, and industry engagement — competences that Dr. Vasquez explicitly requires for her target career trajectory in bio-architecture. The secondment at Holcim Innovation Center adds a third knowledge vector: industrial R&D workflows, scale-up methodologies, and intellectual property management, all critical for translating laboratory findings into deployable products.

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SECTION 2 — IMPACT

2.1 Career Development and Employability of the Researcher

AUMICA is designed as a deliberate career-transformation fellowship. Dr. Vasquez enters with a strong and internationally recognised foundation in environmental metagenomics and synthetic biology, evidenced by publications in Nature Microbiology, Environmental Science and Technology, ACS Synthetic Biology, and Microbiome. However, her stated career objective — a tenure-track position at a European research university at the intersection of microbiology and bio-integrated architecture — requires competences that are not yet part of her profile: materials science, architectural prototyping, industry R&D experience, and the ability to lead interdisciplinary research groups bridging life sciences and the built environment. AUMICA directly and systematically addresses each of these gaps.

By the end of the fellowship, Dr. Vasquez will have: led an original, externally-funded interdisciplinary research project from conception to validated output; acquired advanced materials characterisation competences (XRD, SEM-EDS) independently operable in any materials science facility; completed an industry R&D rotation at Holcim, providing direct experience of translational research workflows and patent processes; supervised research staff and students; managed a project with multiple partners and deliverables; communicated research findings to non-specialist audiences through public exhibitions and media; and published in journals spanning microbiology, materials science, and architecture — demonstrating the cross-disciplinary profile that distinguishes candidates for bio-architecture positions. This combination is currently rare in European academia and positions Dr. Vasquez as a distinctive candidate for emerging faculty roles in sustainable construction, urban ecology, and bio-integrated design.

The Personalised Career Development Plan (PCDP) is jointly designed with Prof. Bellini and will be submitted as a formal deliverable by M3. It maps each training activity and research milestone to a specific competence gap and career objective. Progress against the PCDP will be reviewed formally at M6, M12, and M18 in structured meetings between Dr. Vasquez, Prof. Bellini, and the Politecnico di Milano Career Development Office. The fellowship will be registered on EURAXESS, and Dr. Vasquez will engage with the Politecnico di Milano postdoctoral community network. The non-academic secondment at Holcim additionally enhances intersectoral employability, opening pathways to industrial research careers as an alternative or parallel option. At the European level, AUMICA contributes to the MSCA goal of fostering excellent researchers capable of addressing grand societal challenges — in this case, climate change resilience in the built environment.

2.2 Dissemination, Communication and Exploitation

AUMICA's dissemination and exploitation strategy is structured across three audiences — the scientific community, the construction industry, and the general public — and is designed to maximise reach, reuse, and long-term impact of project outputs.

Scientific dissemination will be delivered through a minimum of three high-impact peer-reviewed publications: (i) the European Building Microbiome Atlas (target: Nature Sustainability or ISME Journal, M10); (ii) design and climate-chamber validation of the three bioresponsive coatings (target: Advanced Materials or ACS Nano, M18); and (iii) the MicroBuild computational model with scenario analysis (target: Building and Environment or npj Computational Materials, M23). All manuscripts will be submitted with open access under a CC BY licence, consistent with Horizon Europe mandates. Preprints will be posted on bioRxiv or EarthArXiv prior to submission to maximise early circulation. Dr. Vasquez will present at two international conferences: the Bio-inspired Materials Congress 2026 (oral presentation, WP2 results) and the World Sustainable Built Environment Conference 2027 (keynote application, full project results). Results will additionally be shared through dedicated

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contributions to the MSCA Alumni Network and the European Association of Research Technology Organisations (EARTO) platforms.

Industry and policy exploitation will be pursued through two primary channels. First, a joint patent application covering the bioresponsive coating formulation will be filed with Politecnico di Milano's Technology Transfer Office and Holcim by M24, protecting the intellectual property emerging from WP2 while enabling commercial scale-up. The Holcim secondment relationship provides a direct pathway to industrial uptake. Second, Dr. Vasquez will engage with the European Construction Technology Platform (ECTP) to present AUMICA results to industry stakeholders and policy audiences, contributing to the update of European standards on bio-integrated building materials.

Public communication will be delivered through two flagship activities: a public installation at Milan Design Week (Salone del Mobile, M22) presenting AUMICA's living facade concept through interactive physical models and real-time microbiome visualisation; and a podcast series titled Buildings That Breathe (six episodes, M10–M20) co-produced with Politecnico di Milano's science communication office, targeting an urban sustainability-interested general audience and distributed via major podcast platforms. Both activities are designed for broad civic engagement and align with the MSCA expectation that researchers engage society with research outcomes. The project website (launched M1) will provide ongoing updates in English and Italian throughout the fellowship duration.

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SECTION 3 — IMPLEMENTATION

3.1 Work Plan

WP	Title	Months	PM	Description
WP1	Urban Microbiome Mapping	M1-M8	8	WP1 generates the empirical foundation for the entire project. Fifty building sites across five European climate zones will be sampled using standardised surface swab and tape-strip protocols. Shotgun
WP2	Bioresponsive Coating Design and Engineering	M6-M16	10	WP2 designs, engineers, and validates three bioresponsive surface coatings using CRISPR-based synthetic biology and cement-compatible biocompatible matrices. Design-build-test-learn cycles will be imp
WP3	Living Facade Validation	M14-M24	9	WP3 deploys the three engineered coatings on Politecnico di Milano's outdoor living building facade testbed for a full 12-month real-world monitoring cycle. Weekly microbiome profiling, continuous env
WP4	Computational Modelling — MicroBuild	M4-M22	10	WP4 develops MicroBuild, an open-source hybrid agent-based and ODE computational model predicting microbiome-material interactions under current and future climate scenarios. The model is calibrated i
WP5	Training, Dissemination and Project Management	M1-M24	5	WP5 encompasses the implementation of the Personalised Career Development Plan, all scientific and public dissemination activities, and overall project management and reporting. It ensures that AUMICA

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WP1: Urban Microbiome Mapping

WP1 generates the empirical foundation for the entire project. Fifty building sites across five European climate zones will be sampled using standardised surface swab and tape-strip protocols. Shotgun metagenomic sequencing at 10 Gb per sample will be performed on all 500 samples. Bioinformatic processing will use the MGSA pipeline for quality filtering, assembly, binning, taxonomic classification, and functional annotation. Culturomics enrichment will be applied to samples from extreme environments. The final deliverable is the open-access European Building Microbiome Atlas deposited in ENA.

T1.1: Sampling site selection and field campaigns — Define 50 building sites across oceanic, Mediterranean, continental, semi-arid, and alpine zones. Conduct standardised surface sampling with full metadata capture including building age, material, orientation, urban zoning, and proximity to land use types relevant to gender analysis.

T1.2: Metagenomic sequencing and bioinformatic processing — Extract DNA, prepare sequencing libraries, and sequence all 500 samples on Illumina NovaSeq 6000. Run MGSA bioinformatic pipeline for taxonomic and functional profiling. Perform statistical analyses including climate zone comparisons, diversity metrics, and gendered land use correlations.

T1.3: Culturomics and biobank establishment — Apply high-throughput culturomics to selected samples to recover culturable taxa. Establish a biobank of 200 functionally characterised surface isolates for use in WP2 consortium design.

ID	Deliverable	Due
D1.1	European Building Microbiome Atlas (dataset deposited in ENA)	M8
D1.2	Scientific publication: metagenomic atlas manuscript submitted	M9

ID	Milestone	Due	Verification
MS1	Microbiome atlas of European building surfaces completed	M8	ENA accession number confirmed; atlas summary report approved by supervisor

WP2: Bioresponsive Coating Design and Engineering

WP2 designs, engineers, and validates three bioresponsive surface coatings using CRISPR-based synthetic biology and cement-compatible biocompatible matrices. Design-build-test-learn cycles will be implemented for each coating type. Climate chamber testing under standardised protocols will confirm performance and durability prior to outdoor deployment.

T2.1: Consortium design and CRISPR engineering — Select chassis organisms from the WP1 biobank. Design and introduce functional genetic modules using CRISPR-Cas12a. Characterise engineered consortia for target function (self-cleaning, NOx metabolism, evapotranspiration), genetic stability, and surface colonisation capacity.

T2.2: Coating formulation with Holcim (secondment M9-M11) — Collaborate with Holcim Innovation Center to develop three biocompatible cement-polymer matrices hosting the engineered consortia. Optimise for microbial viability, adhesion strength, and cement compatibility. Document formulations for patent application.

T2.3: Climate chamber validation — Test all three coatings under controlled climate chamber conditions. Evaluate functional performance, durability under accelerated aging (ISO 16474), and microbiome stability across temperature and humidity gradients.

ID	Deliverable	Due
D2.1	Three bioresponsive coating prototypes characterised and documented	M16
D2.2	Scientific publication: coating design and climate chamber validation submitted	M18

ID	Milestone	Due	Verification
MS2	Three bioresponsive coatings validated in	M16	Performance metrics (contact angle, NOx

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	climate chambers		removal rate, temperature differential) meet pre-defined thresholds in climate chamber tests

WP3: Living Facade Validation

WP3 deploys the three engineered coatings on Politecnico di Milano's outdoor living building facade testbed for a full 12-month real-world monitoring cycle. Weekly microbiome profiling, continuous environmental sensing, and monthly functional assays will provide a comprehensive performance dataset spanning all four seasons.

T3.1: Facade preparation and coating application — Prepare designated facade panels and apply the three coatings according to standardised protocols developed in WP2. Install environmental sensor arrays (temperature, humidity, particulate matter, NOx) and establish data logging infrastructure.

T3.2: 12-month real-world monitoring programme — Conduct weekly surface microbiome sampling (16S rRNA and ITS amplicon sequencing), monthly functional performance assays, and continuous sensor data collection. Analyse temporal dynamics of microbiome community structure and functional output under natural seasonal variation.

T3.3: Data integration and validation reporting — Integrate microbiome, functional, and environmental sensor datasets. Assess coating performance against pre-defined success criteria. Prepare validation report and patent filing documentation in coordination with Holcim.

ID	Deliverable	Due	
D3.1	12-month facade monitoring dataset (deposited open access)	M24	
D3.2	Facade validation report and patent application filed	M24	
ID	**Milestone**	**Due**	**Verification**
MS4	Facade validation report completed and patent application filed	M24	Validation report approved by supervisor and Technology Transfer Office; patent application receipt confirmed

WP4: Computational Modelling — MicroBuild

WP4 develops MicroBuild, an open-source hybrid agent-based and ODE computational model predicting microbiome-material interactions under current and future climate scenarios. The model is calibrated iteratively against WP1 and WP3 empirical data and released publicly under MIT licence.

T4.1: Model architecture design and initial development — Design model architecture integrating microbial growth kinetics, substrate diffusion, climate forcing, and material degradation functions. Develop initial codebase in Python on public GitHub repository. Release alpha version for community feedback at M10.

T4.2: Calibration with empirical data — Calibrate model parameters using WP1 Atlas data (M8 onwards) and WP3 real-world monitoring data (M16 onwards). Perform sensitivity analysis and uncertainty quantification.

T4.3: Climate scenario simulations and MicroBuild v1.0 release — Run scenario simulations under RCP 4.5 and RCP 8.5 climate trajectories for five European cities. Finalise documentation, testing suite, and user guide. Release MicroBuild v1.0 on GitHub and Zenodo with DOI by M22.

ID	Deliverable	Due	
D4.1	MicroBuild v1.0 released on GitHub and Zenodo	M22	
D4.2	Scientific publication: MicroBuild model and scenario analysis submitted	M23	
ID	**Milestone**	**Due**	**Verification**
MS3	MicroBuild v1.0 publicly released	M22	GitHub release tag confirmed; Zenodo DOI issued; model passes automated test suite

WP5: Training, Dissemination and Project Management

WP5 encompasses the implementation of the Personalised Career Development Plan, all scientific

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and public dissemination activities, and overall project management and reporting. It ensures that AUMICA delivers on its training, outreach, and administrative commitments throughout the fellowship.

T5.1: PCDP implementation and training activities — Implement all training activities defined in the PCDP including the Project Management course (M2), Materials Characterisation training (M3 onwards), Science Communication workshop (M4), Italian language course (M1-M12), and Master's student supervision (M6-M18). Formal PCDP reviews at M6, M12, M18.

T5.2: Dissemination and public communication — Manage project website and social media. Produce Buildings That Breathe podcast series (M10-M20). Prepare and present conference contributions (M16, M23). Coordinate Milan Design Week installation (M22). Manage all open-access publication submissions.

T5.3: Project management and reporting — Maintain project monitoring against the work plan. Prepare mid-term and final progress reports for the European Commission. Coordinate with Holcim for secondment logistics. Manage ethical compliance and data management plan updates.

ID	Deliverable	Due
D5.1	Data Management Plan v1.0	M6
D5.2	Personalised Career Development Plan	M3
D5.3	Mid-term progress report	M12
D5.4	Final project report	M24

ID	Milestone	Due	Verification
MS0	Project kick-off and PCDP finalised	M3	PCDP document approved by supervisor and submitted to EC portal

Summary of Deliverables

ID	Title	WP	Due	Type
D1.1	European Building Microbiome Atlas (dataset deposited in ENA)	WP1	M8	Open dataset
D1.2	Scientific publication: metagenomic atlas manuscript submitted	WP1	M9	Publication
D2.1	Three bioresponsive coating prototypes characterised and documented	WP2	M16	Technical report
D2.2	Scientific publication: coating design and climate chamber validation submitted	WP2	M18	Publication
D3.1	12-month facade monitoring dataset (deposited open access)	WP3	M24	Open dataset
D3.2	Facade validation report and patent application filed	WP3	M24	Technical report and patent filing
D4.1	MicroBuild v1.0 released on GitHub and Zenodo	WP4	M22	Open-source software
D4.2	Scientific publication: MicroBuild model and scenario analysis submitted	WP4	M23	Publication
D5.1	Data Management	WP5	M6	Data management

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	Plan v1.0			plan
D5.2	Personalised Career Development Plan	WP5	M3	Management document
D5.3	Mid-term progress report	WP5	M12	Progress report
D5.4	Final project report	WP5	M24	Final report

Summary of Milestones

ID	Title	WP	Due	Verification
MS1	Microbiome atlas of European building surfaces completed	WP1	M8	ENA accession number confirmed; atlas summary report approved by supervisor
MS2	Three bioresponsive coatings validated in climate chambers	WP2	M16	Performance metrics (contact angle, NOx removal rate, temperature differential) meet pre-defined thresholds in climate chamber tests
MS4	Facade validation report completed and patent application filed	WP3	M24	Validation report approved by supervisor and Technology Transfer Office; patent application receipt confirmed
MS3	MicroBuild v1.0 publicly released	WP4	M22	GitHub release tag confirmed; Zenodo DOI issued; model passes automated test suite
MS0	Project kick-off and PCDP finalised	WP5	M3	PCDP document approved by supervisor and submitted to EC portal

Work Plan Timeline

Work Package	M1-M6	M7-M12	M13-M18	M19-M24
WP1	[redacted]	[redacted]		
WP2	[redacted]	[redacted]	[redacted]	
WP3			[redacted]	[redacted]
WP4	[redacted]	[redacted]	[redacted]	[redacted]
WP5	[redacted]	[redacted]	[redacted]	[redacted]

WP1 (Urban Microbiome Mapping) runs M1-M8, with field campaigns in M1-M5 and metagenomic analysis in M4-M8. WP2 (Coating Design) runs M6-M16, with consortium engineering in M6-M10, the Holcim secondment in M9-M11, and climate chamber validation in M11-M16. WP3 (Living Facade Validation) runs M14-M24, with coating application in M14-M15 and continuous monitoring in M15-M24. WP4 (Computational Modelling) runs M4-M22, with model development in M4-M10, calibration in M8-M18, and scenario simulations in M18-M22. WP5 (Training, Dissemination, Management) runs continuously M1-M24. Key overlaps ensure empirical data from WP1 feeds WP2 consortium design from M6, WP3 validation overlaps with WP4 calibration from M15, and all research WPs are supported by WP5 throughout.

3.2 Management Structure and Risk Management

Project management will be conducted under a straightforward bilateral structure between Dr. Vasquez (fellow and day-to-day project manager) and Prof. Bellini (supervisor and institutional

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contact). Monthly one-to-one progress meetings will be held between Dr. Vasquez and Prof. Bellini, supplemented by bi-monthly group meetings with the BAL research team. A formal Project Steering Group meeting will be held at M6, M12, and M18, including a representative from Holcim as associated partner. Dr. Vasquez will maintain a live project management log on a shared platform (Notion or equivalent) tracking deliverable status, milestone progress, and risk indicators. The Politecnico di Milano Research Services Office will provide administrative support for financial reporting and ethical compliance monitoring.

Risk assessment and mitigation: Three principal risks have been identified. Risk 1 — Low microbial diversity on building surfaces in certain climate zones (probability: medium; impact: high). If surface microbiome diversity in specific zones proves lower than anticipated, the taxonomic basis for consortium design could be constrained. Mitigation: the sampling design deliberately includes both aged and newly constructed buildings, as building age is a key predictor of microbial richness. Culturomics enrichment protocols will be applied to all low-diversity samples to recover viable taxa not detectable by metagenomics. Additionally, the consortium design in WP2 can draw on well-characterised laboratory strains as chassis organisms if environmental isolates prove insufficiently robust. Risk 2 — Coating durability failures under extreme weather conditions (probability: low-medium; impact: high). If outdoor facade coatings degrade beyond acceptable performance thresholds under real-world weather extremes, the WP3 validation results could be compromised. Mitigation: accelerated aging tests in climate chambers (ISO 16474, 500-hour UV-condensation cycles) are mandatory gates before facade deployment. Only coatings passing defined performance thresholds in WP2 will proceed to WP3. Formulation redundancy is built in — three coating types are engineered, so partial failure of one does not invalidate the facade validation programme. Risk 3 — Holcim secondment scheduling delays (probability: low; impact: medium). Industrial partner scheduling constraints could delay the M9-M11 secondment window. Mitigation: the secondment window has contractual flexibility of plus or minus two months, shifting it to M8-M10 or M10-M12 without affecting WP2 downstream timelines. A remote collaboration protocol with Holcim has been agreed, allowing preliminary formulation work to proceed by video conference and materials courier if on-site access is delayed.

3.3 Institutional Environment and Hosting Arrangements

Politecnico di Milano is Italy's leading technical university and one of Europe's foremost research institutions in engineering, architecture, and design, consistently ranked in the global top 20 for Architecture and Built Environment (QS World University Rankings 2024: rank 3). The Department of Architecture, Built Environment and Construction Engineering (ABC) is the largest and most research-intensive architecture and construction department in Italy, with annual research expenditure exceeding EUR 15 million and active participation in over 40 funded research projects. The department hosts Europe's only dedicated living building facade testbed — a purpose-built outdoor experimental facility equipped with modular panel systems, environmental sensor networks, and real-time data logging — which is the irreplaceable experimental infrastructure for AUMICA's WP3 validation programme.

The Bio-Architecture Laboratory (BAL), directed by Prof. Bellini, occupies 450 m2 of dedicated laboratory and prototyping space within the ABC department. It is equipped with climate simulation chambers (temperature range -20°C to +60°C, humidity 10-95% RH), materials characterisation instruments (XRD, SEM-EDS, confocal microscopy, contact angle goniometer, UV-Vis spectrometer), a biological safety level 1 wet laboratory with standard molecular biology equipment, and a full-scale architectural prototyping workshop. These facilities comprehensively support all experimental activities of WP1 through WP3. For metagenomic sequencing, AUMICA will use the Next-Generation Sequencing Core at Politecnico di Milano's affiliated IRCCS San Raffaele Research Hospital, with which Prof. Bellini has an established collaboration agreement providing preferential access and cost arrangements.

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The Politecnico di Milano Research Services Office provides dedicated administrative support for EU-funded projects, including a team of experienced project managers specialised in Horizon Europe reporting requirements. The university operates an MSCA Support Service offering fellows institutional onboarding, career development resources, and peer networking. Holcim Innovation Center (Zurich, Switzerland) joins as an associated partner for the secondment, contributing industrial R&D infrastructure, proprietary cement formulation expertise, and a direct route to commercial exploitation of WP2 outputs. The institutional environment is thus not merely adequate — it is uniquely suited to AUMICA's interdisciplinary ambition.

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PART B2

SECTION 4 — CV of the Experienced Researcher

Education

Dr. Elena Vasquez holds a PhD in Environmental Microbiology awarded in March 2022 by ETH Zurich, Switzerland, one of the world's leading technical universities. Her doctoral dissertation, entitled 'Metagenomic Approaches to Understanding Microbial Colonisation Dynamics on Engineered Urban Surfaces,' was supervised by Prof. Klaus Zimmermann and received the ETH Medal for Outstanding Doctoral Thesis. Prior to her doctorate, she completed a Master of Science in Environmental Engineering (distinction) at the University of Barcelona (2017) and a Bachelor of Science in Biology at the Universidad Complutense de Madrid (2015).

Research Experience

Dr. Vasquez is currently a Postdoctoral Research Fellow at Wageningen University and Research (Netherlands) in the Laboratory of Microbiology, a position she has held since October 2022. In this role she leads an independent research line on biofilm engineering for environmental applications, manages a team of two PhD students and one research technician, and holds co-investigator status on a NWO-funded project on microbial carbon capture. During her doctoral work at ETH Zurich (2017-2022) she developed the bioinformatic pipeline MGSA for metagenomic analysis of surface-associated microbial communities, which has been adopted by three independent research groups. She has accumulated expertise across environmental metagenomics, culturomics, CRISPR-based synthetic biology, agent-based computational modelling, and urban ecology, spanning the full experimental-computational spectrum required for AUMICA. She has three years of postdoctoral experience, within the MSCA-PF eligibility window, and no mobility to Italy in the 12 months preceding the call deadline, satisfying the MSCA mobility rule.

Publications

1. Vasquez E., Muller T., Hoffmann A., Zimmermann K. (2024). Metagenomic profiling of building-surface microbiomes across European climate zones. Nature Microbiology, 12(3), 445-458. https://doi.org/10.1038/s41564-024-01XXX
2. Vasquez E., Lindstrom K. (2023). Engineered biofilms for atmospheric CO2 capture: a proof-of-concept. Environmental Science and Technology, 57(8), 3201-3215. https://doi.org/10.1021/acs.est.3cXXXXX
3. Vasquez E., De Groot R., Patel S., Zimmermann K. (2023). CRISPR-guided microbial consortia for self-healing concrete. ACS Synthetic Biology, 12(1), 89-102. https://doi.org/10.1021/acssynbio.2cXXXXX
4. Vasquez E., Chen W. (2022). Urban microbiome dynamics under heat stress: implications for building material degradation. Microbiome, 10, Article 178. https://doi.org/10.1186/s40168-022-XXXXX

H-index: 9 (Google Scholar, as of submission date). Total citations: 312. ORCID: 0000-0001-XXXX-XXXX.

Other Achievements

Awards and recognition: ETH Medal for Outstanding Doctoral Thesis (2022); Best Poster Prize, International Society for Microbial Ecology (ISME) Conference, Lausanne (2021); ETH Zurich Excellence Scholarship (2017-2022). Invited contributions: invited speaker at the European Microbiome Congress (2023); invited reviewer for Nature Microbiology, Environmental Microbiology, and Biofouling. Supervision and teaching: co-supervision of two PhD students at Wageningen University; teaching assistant for Environmental Biotechnology MSc module (2023-2024). Science communication: contributor to the science blog MicrobeWorld (3 articles, 2022-2023); interview on Dutch national radio NPO1 on urban microbiome research (2023). Mobility: Dutch residence (2022-

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present), Swiss residence (2017-2022), Spanish origin. No residence or main activity in Italy in the 12 months prior to the call deadline, satisfying the MSCA-PF mobility eligibility criterion.

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SECTION 5 — Capacity of the Participating Organisations

Host Organisation Capacity

Politecnico di Milano (PoliMi) is a public technical university founded in 1863 and is Italy's largest and most research-intensive technical university, with approximately 47,000 students and 1,500 faculty members. It is consistently ranked among the world's top institutions in Engineering and Technology (QS World University Rankings 2024: rank 139 overall; rank 3 in Architecture and Built Environment). PoliMi has extensive experience in managing European research funding, having participated in over 1,200 EU-funded projects and coordinated more than 150 Horizon Europe grants. The university's dedicated EU Project Office provides professional grant management support, financial reporting, and compliance monitoring for all EU-funded projects. PoliMi is a signatory of the European Charter for Researchers and the Code of Conduct for the Recruitment of Researchers, implementing open, merit-based, and transparent researcher recruitment procedures. The university's commitment to researcher career development is demonstrated through its dedicated MSCA Support Service, which provides fellows with institutional onboarding, mentoring, career development workshops, and community networking throughout the fellowship period. PoliMi has hosted 23 MSCA Individual Fellowships and Postdoctoral Fellowships since 2014, with a completion rate of 100% and a publication output rate above the MSCA average.

Supervisor Profile

Prof. Marco Bellini is Full Professor of Sustainable Building Technologies at the Department of Architecture, Built Environment and Construction Engineering (ABC) at Politecnico di Milano, a position he has held since 2014. He holds a PhD in Building Engineering from Politecnico di Milano (2001) and has held visiting positions at TU Delft and ETH Zurich. His research focuses on bio-integrated facade systems, responsive building materials, circular construction, and the integration of biological processes into the built environment. He has authored over 120 peer-reviewed publications (h-index: 24), holds 4 patents on bio-integrated building components, and has coordinated three major EU-funded projects: BioFacade (FP7), CircularBuild (H2020), and GreenSkin (Horizon Europe, ongoing). Prof. Bellini has supervised 18 PhD students to successful completion and currently supervises 6 PhD students and 4 postdoctoral researchers. He has previously hosted 3 MSCA fellows and is familiar with the administrative, supervisory, and reporting requirements of the programme. His expertise in architectural design, building physics, and materials engineering is precisely complementary to Dr. Vasquez's profile in environmental microbiology and synthetic biology, creating the interdisciplinary synthesis that is the scientific basis of AUMICA.

Infrastructure and Resources

The Bio-Architecture Laboratory (BAL) at Politecnico di Milano ABC department provides the full range of infrastructure required for AUMICA. Research infrastructure includes: (i) climate simulation chambers capable of reproducing temperature conditions from -20 to +60 degrees Celsius and relative humidity from 10 to 95%, with programmable dynamic cycling for accelerated aging experiments; (ii) materials characterisation suite including X-ray diffractometer (XRD), scanning electron microscope with energy-dispersive X-ray spectroscopy (SEM-EDS), confocal laser scanning microscope, contact angle goniometer (sessile drop), and UV-Vis spectrophotometer; (iii) biological safety level 1 wet laboratory with standard molecular biology equipment (thermocyclers, gel electrophoresis, plate readers, anaerobic workstation, -80 degrees Celsius storage); (iv) full-scale architectural prototyping workshop with CNC machining, 3D printing, and composite fabrication capability. The department's living building facade testbed — an outdoor experimental facility on the ABC department building comprising 48 interchangeable modular panels (each 0.5m x 1.0m), a continuous environmental sensor network (temperature, humidity, PAR, NOx, PM2.5), and real-time data logging at 1-minute intervals — is the irreplaceable experimental platform for AUMICA's WP3 validation. Sequencing services are provided through a formal collaboration agreement with the NGS Core at IRCCS San Raffaele Research Hospital, accessible at preferential rates.

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SECTION 6 — Ethical Aspects

AUMICA does not involve research on human participants, human cells, human tissues, or personal data. No clinical research or intervention is planned. The project will not collect or process personal data as defined under the General Data Protection Regulation (EU) 2016/679, and therefore no specific data protection measures beyond standard information security practices for research data are required. There are no activities involving animals. No research activities will be conducted in third countries outside the EU or associated countries; the Holcim secondment is located in Switzerland, an MSCA-associated country, and does not raise third-country ethical concerns. All field sampling in WP1 is conducted on building surfaces in publicly accessible urban environments or with the agreement of building owners, and does not involve collection of personal or sensitive information.

Genetically modified organisms (GMO) and biosafety: WP2 involves CRISPR-Cas12a-based genetic modification of bacterial chassis organisms under biosafety level 1 (BSL-1) containment conditions in the BAL wet laboratory at Politecnico di Milano. The organisms selected as chassis will be Risk Group 1 microorganisms (non-pathogenic, non-toxigenic, with negligible risk to healthy adults) in accordance with EU Directive 2009/41/EC on contained use of genetically modified micro-organisms. All GMO work will be carried out under a contained use notification submitted to the competent Italian authority (Ministry of Health) prior to the start of WP2 activities. No deliberate environmental release of GMOs is planned during the fellowship; the WP3 living facade validation uses bioresponsive coatings in which microbial viability is retained within the coating matrix and evaluated under natural but contained conditions. An environmental risk assessment for the facade deployment will be conducted in consultation with Politecnico di Milano's Biosafety Committee prior to WP3 commencement. No dual-use concerns have been identified for the technologies developed in AUMICA. The project will be conducted in full compliance with the European Code of Conduct for Research Integrity (ALLEA, 2023).

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